

GLS, ANCOVA, and Averaged Change-Score Estimators for Clinical Trials with Run-In Observations

Ronald G. Thomas, Ph.D.*

September 07, 2026

Abstract

Background. Three estimators of the treatment effect are in routine use for longitudinal trials with run-in observations. The GLS estimator, under the linear mixed model, exploits the full covariance structure to extract maximum information from all observations. The ANCOVA estimator regresses post-randomization means on the run-in mean as a covariate, without specifying the random-effects covariance. The averaged change-score estimator computes a simple difference of phase means and applies a t -test, requiring no variance-component estimation. To the best of our knowledge, the relative efficiency of these estimators under correct and misspecified models has not been comprehensively characterized. **Methods.** We derive the variance of the three estimators as a function of the trial design (J_0, J_1, J_2) and the variance parameters (σ^2, σ_b^2) . The derivation proceeds via the Woodbury-based variance formulas of Paper 1 in this compendium, plus closed-form expressions for the ANCOVA and averaged-change-score variances under the same random-slopes model. We assess robustness to misspecification of the within-subject covariance (R) and the random-effects covariance (G) analytically, through sandwich variances. We also run a Monte Carlo simulation under compound symmetry that evaluates the estimators alongside a GLS analysis that misstates the random-slope variance. **Results.** We make all comparisons on the common estimand scale, rescaling the averaged and ANCOVA contrasts to unbiasedness for the treatment effect γ . Under correct specification, we find $\text{GLS} \leq \text{ANCOVA} \leq \text{averaged}$ in variance, with no violations over a 115-cell grid of designs and signal-to-noise ratios. The efficiency penalty of the simpler estimators is substantial in run-in designs with multiple treatment-period visits. At $(J_0, J_1, J_2) = (2, 2, 0)$, the relative efficiency of ANCOVA versus GLS ranges from 0.64 to 0.81 across signal-to-noise ratios, an efficiency loss of 19% to 36%. Relative efficiency above 0.95 occurs only in single-post-visit run-in designs ($J_1 = 1, J_2 = 0$), and in designs without run-in or common close ($J_0 = 0, J_2 = 0$) at high signal-to-noise. In the MIRIAD-based example, the ANCOVA sample size exceeds the GLS sample size by up to a factor of roughly 2.6 among run-in designs, and the averaged sample size by up to a factor of roughly 45. Under the G and R misspecifications studied, GLS never loses its advantage over the simpler estimators. That is, the misspecified-GLS sandwich variance peaks at 5.3, against an averaged-estimator variance of 8.9 at the reference design. **Conclusions.** We conclude that GLS is the efficient estimator under correct specification, and that its advantage survives the covariance misspecifications examined here. The averaged change-score estimator remains useful as a fully prespecified robust benchmark and a teaching tool. One open robustness question, which we do not study here, is whether feasible GLS with variance components estimated from small samples retains this advantage.

*Wertheim School of Public Health, UCSD; ORCID: 0000-0003-1686-4965

41	Contents	
42	1 Introduction	2
43	2 Three estimators	4
44	2.1 GLS under the mixed model	4
45	2.2 Averaged change score (<i>t</i> -test)	4
46	2.3 ANCOVA with run-in mean as covariate	5
47	2.4 SUR (seemingly unrelated regression)	6
48	2.5 BLUP as theoretical benchmark	6
49	3 Relative efficiency under correct specification	7
50	3.1 Analytical comparison	7
51	3.2 When does the gap exceed 5%?	8
52	4 Robustness to misspecification	10
53	4.1 Misspecified <i>G</i>	10
54	4.2 Misspecified <i>R</i>	11
55	5 Simulation study	11
56	5.1 Design	11
57	5.2 Results	12
58	6 Numerical examples	13
59	6.1 MIRIAD parameters	13
60	6.2 When to use which estimator	13
61	7 Discussion	13
62	8 Conflict of interest	14
63	9 Funding	14
64	10 Data availability statement	14
65	11 Reproducibility	15
66	12 References	15

67 1 Introduction

68 The GLS estimator $\hat{\gamma}_{\text{GLS}} = (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}\mathbf{c}$ is the best linear unbiased estimator (mini-
69 mum variance among linear unbiased estimators) when Σ is known. In practice, however, Σ
70 must be estimated, and the efficiency of the feasible GLS estimator depends on the accuracy
71 of that estimation. With small sample sizes or short follow-up, variance component estimates
72 can be unstable, and this instability may erode the theoretical advantage of GLS.

73 The averaged change-score approach, by contrast, sidesteps variance component estimation

entirely. For a trial with J_0 run-in and J_2 common close observations, the estimator is:

$$\hat{\gamma}_{\text{avg}} = \bar{d}_1 - \bar{d}_0 \quad (1)$$

where

$$d_{ik} = \frac{1}{J_1 + J_2} \sum_{\ell=1}^{J_1+J_2} Y_{i,\ell,k} - \frac{1}{J_0} \sum_{\ell=1}^{J_0} Y_{i,-\ell,k} \quad (2)$$

is the difference of the post-randomization and run-in means for participant i . We note that the baseline visit is the change-score reference and cancels, so it enters neither average; this matches the contrast implemented in `var_gamma_avg()`. Throughout, the code and figure parameters use (r, k, f) for the design counts, with the correspondence $J_0 = r$ (run-in), $J_1 = k$ (treatment period), and $J_2 = f$ (common close). The prose and table headers, on the other hand, use (J_0, J_1, J_2) .

The debate between change-score and ANCOVA analyses has a long history^{1-3,4} provided a formal efficiency comparison for pretest-posttest designs. The closest antecedent to the present work is⁵, who derived the relative efficiency of post-treatment, change-score, and ANCOVA analyses when several baseline measurements are available, and⁶, who extended the treatment to repeated post-treatment measurements. Their ordering, that ANCOVA on the mean of the available baselines dominates the change score, which in turn dominates the raw post-treatment comparison, is the qualitative result that the ordering established in Section 3.1 specializes. What is new here is not the ordering itself but the closed-form relative efficiencies in the run-in random-slope parameterization. Specifically, the efficiency margins are obtained as explicit functions of the design counts (J_0, J_1, J_2) and the signal-to-noise ratio σ_b/σ ; the analysis covers a post-treatment common close phase absent from the Frison-Pocock setting; and the estimand is a difference of slopes rather than a difference of means. Within the run-in literature,⁷ derived GLS power formulas for trials with pre-randomization observations, and we take this as the efficient benchmark against which the simpler estimators are measured.⁸ and⁹ compared baseline-as-covariate against baseline-as-response (constrained longitudinal data analysis).¹⁰ developed a semiparametric framework for covariate adjustment.¹¹ showed that MMRM requires time-by-covariate interactions to guarantee power gains over ANCOVA.¹² established the foundational comparison of pretest-posttest designs.¹³ addressed conditioning on baseline in longitudinal RCTs.¹⁴ proposed the two-period approach for run-in trials.

We note that the averaged change-score estimator studied here is distinct from (and less efficient than) the constrained longitudinal data analysis (cLDA) estimator of¹⁵ and⁹, which includes baseline in the response vector with a common-mean constraint. The cLDA estimator uses the full covariance structure without requiring specification of the random effects distribution G , placing it between the averaged estimator and GLS in the efficiency ordering.¹⁶ extended the cLDA framework to pre- and post-treatment measurements with equal baseline assumptions. A full comparison including cLDA is beyond the scope of this paper, but it represents a natural extension of the present work.

The `longpower` R package¹⁷ provides the CRAN reference implementation of power calculations for the GLS / REML-based slope-comparison estimator, covering the Liu-Liang¹⁸,

113 Diggle¹⁹, and Edland²⁰ formulations. These routines correspond to the $\hat{\gamma}_{\text{GLS}}$ estimator stud-
 114 ied here under correct specification and constitute the Paper 1 baseline, extended in this
 115 paper by run-in and common close observations. However, `longpower` does not cover the
 116 ANCOVA-on-run-in-mean estimator or the averaged change-score t -test, nor does it charac-
 117 terize the efficiency loss from misspecification of G or R . The comparison developed here
 118 therefore complements `longpower` in two ways: (i) it orders the three candidate estimators
 119 under correct specification, and (ii) it quantifies the efficiency margin that GLS retains under
 120 the misspecifications of G and R studied below.

121 2 Three estimators

122 2.1 GLS under the mixed model

123 We begin with the GLS estimator. From²¹, the per-pair variance is the (γ, γ) element of
 124 $(X'\Sigma^{-1}X)^{-1}$ with $\Sigma = R + ZGZ'^{22,23}$, computed via the Woodbury identity.

125 For the Frost conventional design ($J_0 = 0$, $J_1 = 2$):

$$126 \quad \hat{\gamma}_{\text{GLS}} = \frac{1}{2t}[(C_{22} + C_{21}) - (C_{12} + C_{11})] \quad (3)$$

127 with $\text{Var}(\hat{\gamma}_{\text{GLS}}) = (\sigma^2 + 2\sigma_b^2 t^2)/t^2$.

128 2.2 Averaged change score (t -test)

129 The averaged change-score estimator ignores the within-subject correlation structure and
 130 treats d_{ik} as independent observations:

$$131 \quad \text{Var}(\hat{\gamma}_{\text{avg}}) = \frac{2}{m} \text{Var}(d_{ik}) \quad (4)$$

132 The variance $\text{Var}(d_{ik})$ depends on the correlation structure, but its estimation does not require
 133 the correlation structure to be specified. To obtain an estimator of γ , we divide the contrast
 134 by the design constant c derived in Section 3.1. All variances, relative efficiencies, and sample
 135 sizes reported in this paper are on that common γ scale.

```
a <- 1.0
b <- 1.0
tt <- 1.0

# gamma-scale rescaling constant (Section 3.1):
# mean accumulated effect over post-randomization visits
cfac <- function(r, k, f, t_interval = 1) {
  t_interval * (k * (k + 1) / 2 + k * f) / (k + f)
}

v_gls <- var_gamma_matrix(2, 2, 0, a, b, tt)
```

```

v_avg <- var_gamma_avg(2, 2, 0, a, b, tt) /
  cfac(2, 2, 0, tt)^2
re <- v_gls / v_avg

cat(sprintf('J_0=2, J_1=2: GLS=%.4f, Avg=%.4f, RE=%.4f\n',
            v_gls, v_avg, re))

```

```
136 ## J_0=2, J_1=2: GLS=2.1569, Avg=8.8889, RE=0.2426
```

137 2.3 ANCOVA with run-in mean as covariate

138 The ANCOVA approach, in contrast, uses the averaged run-in rate as a baseline covariate
 139 without specifying σ_b^2 :

$$140 \bar{Y}_{i,\text{post}} = \alpha + \gamma g_i + \phi \bar{Y}_{i,\text{pre}} + \varepsilon_i \quad (5)$$

141 The variance of $\hat{\gamma}$ under this model is

$$142 \text{Var}(\hat{\gamma}_{\text{ANC}}) = \frac{2}{m} \text{Var}(\bar{Y}_{\text{post}}) (1 - \rho_{\text{pre,post}}^2) \quad (6)$$

143 where $\rho_{\text{pre,post}}$ is the marginal correlation between the run-in and post-randomization means.
 144 When $J_0 = 0$, there is no covariate, and ANCOVA reduces to the averaged estimator. When
 145 $\rho_{\text{pre,post}}$ is large, that is, when σ_b^2 dominates σ^2 , the $(1 - \rho^2)$ factor produces a substantial
 146 reduction in variance.

147 The key advantage of ANCOVA over GLS is that ϕ is estimated from the data rather than
 148 derived from assumed variance components^{1,24}. Its key advantage over the averaged estimator,
 149 on the other hand, is that it exploits the predictive value of the run-in observations^{2,25}.

```

c220 <- cfac(2, 2, 0, tt)
v_gls <- var_gamma_matrix(2, 2, 0, a, b, tt)
v_anc <- var_gamma_ancova(2, 2, 0, a, b, tt) / c220^2
v_avg <- var_gamma_avg(2, 2, 0, a, b, tt) / c220^2

cat(sprintf('J_0=2, J_1=2 (sigma_b/sigma = 1):\n'))

```

```
150 ## J_0=2, J_1=2 (sigma_b/sigma = 1):
```

```
cat(sprintf('  GLS:    %.4f\n', v_gls))
```

```
151 ##    GLS:    2.1569
```

```
cat(sprintf('  ANCOVA: %.4f\n', v_anc))
```

```
152 ##    ANCOVA: 2.9630
```

```
cat(sprintf('  Avg:    %.4f\n', v_avg))
```

```
153 ##    Avg:    8.8889
```

```
cat(sprintf(' RE(GLS/ANC): %.3f\n', v_gls / v_anc))
```

```
## RE(GLS/ANC): 0.728
```

```
cat(sprintf(' RE(ANC/Avg): %.3f\n', v_anc / v_avg))
```

```
## RE(ANC/Avg): 0.333
```

2.4 SUR (seemingly unrelated regression)

The SUR approach treats the run-in and treatment periods as separate regression equations, linked by correlated errors:

$$Y_1 = X_2 \begin{pmatrix} \beta_{11} \\ \beta_{12} \end{pmatrix} + e_1 \quad (\text{both groups}) \quad (7)$$

$$Y_2 = X_1 \alpha_2 + e_2 \quad (\text{baseline}) \quad (8)$$

The deviation-from-mean parameterization gives $\beta_1 = (\delta_1 + \delta_2)/2$ and $\beta_2 = (\delta_2 - \delta_1)/2$, where δ_j are the group-specific rates.

We do not develop the SUR variance derivation or its comparison to the GLS and averaged estimators in this paper. The subsection is retained as a specification only, and we flag it as future work in the Discussion.

2.5 BLUP as theoretical benchmark

The three estimators above are linear functions of the observed change scores, and we can compare them to a common benchmark: the best linear unbiased predictor (BLUP) of the individual random slopes. Under the random-slopes model with marginal covariance $V_i = R_i + Z_i G Z_i'$, the BLUP of the subject-specific random slope b_i given the data is

$$\hat{b}_i = G Z_i' V_i^{-1} (y_i - X_i \hat{\beta}) \quad (9)$$

where $\hat{\beta}$ is the GLS estimator of the fixed effects^{26,27}.

The group-specific slope estimators derived from the fixed effects share the same variance as $(\hat{\beta})_\gamma$, and the GLS estimator $\hat{\gamma}_{\text{GLS}}$ minimizes the variance among all linear unbiased estimators of γ when V_i is correctly specified. We regard the averaged and ANCOVA estimators as specific linear approximations to this benchmark:

- **Averaged change-score** replaces V_i^{-1} with an unweighted sum; this is optimal only when $V_i \propto I$, which fails for any non-zero random-slope variance.
- **ANCOVA** collapses the run-in visits into a single scalar covariate (the run-in mean) rather than weighting them by V_i^{-1} ; the resulting efficiency loss vanishes as $J_0 \rightarrow \infty$ since the run-in mean becomes a sufficient statistic for the subject-specific intercept plus average slope.

183 Section 4 examines whether model misspecification erodes the nominal optimality of GLS
 184 relative to these approximations, which depend on fewer variance components. In the G and
 185 R misspecifications studied there, the GLS variance increases but remains below that of the
 186 averaged estimator. Whether feasible GLS with estimated components loses its advantage in
 187 small samples is a question we do not take up here.

188 3 Relative efficiency under correct specification

189 3.1 Analytical comparison

190 The relative efficiency of the averaged estimator to GLS is:

$$191 \quad \text{RE}_{\text{avg}} = \frac{\text{Var}(\hat{\gamma}_{\text{GLS}})}{\text{Var}(\hat{\gamma}_{\text{avg}}/c)} \leq 1 \quad (10)$$

192 with equality when the averaged estimator happens to coincide with GLS, which occurs only
 193 in special cases.

194 We can state and prove the ordering asserted above directly. One preliminary is needed. As
 195 defined, the averaged and ANCOVA contrasts average the accumulated treatment effect over
 196 the post-randomization visits, so their expectation is $c\gamma$ rather than γ , where

$$197 \quad c = t \frac{J_1(J_1 + 1)/2 + J_1 J_2}{J_1 + J_2} \quad (11)$$

198 is the mean of the accumulated-effect design column over the post-randomization visits ($c =$
 199 $1.5t$ for the $(2, 2, 0)$ design; $c = t$ when $J_1 = 1$, $J_2 = 0$). Dividing either contrast by c therefore
 200 yields an unbiased estimator of γ , with variance scaled by $1/c^2$.

201 **Proposition.** Under the random-slopes model with correctly specified Σ , $\text{Var}(\hat{\gamma}_{\text{GLS}}) \leq$
 202 $\text{Var}(\hat{\gamma}_{\text{ANC}}/c) \leq \text{Var}(\hat{\gamma}_{\text{avg}}/c)$.

203 *Proof.* Each rescaled contrast is a linear unbiased estimator of γ , and $\hat{\gamma}_{\text{GLS}}$ is the best linear
 204 unbiased estimator of γ when Σ is known (Gauss-Markov). This gives the first inequality. For
 205 the second, both contrasts have the form (post-randomization mean) $-\phi \times$ (run-in mean). The
 206 population ANCOVA coefficient minimizes the residual variance over ϕ , while the averaged
 207 estimator fixes $\phi = 1$, so the ANCOVA contrast variance is no larger; dividing both by c^2
 208 preserves the inequality. (The feasible ANCOVA estimator estimates ϕ from the data and
 209 carries the usual finite-sample inflation.) $\square$

210 We emphasize that the ordering itself is not new.⁵ established the same qualitative ranking for
 211 repeated-measures designs with several baseline measurements, and⁶ extended it to repeated
 212 post-treatment measurements. The argument above is the specialization of that result to
 213 the run-in random-slope model, where the run-in mean plays the role of their baseline mean.
 214 The contribution of the following sections is the closed-form magnitude of the two gaps as
 215 functions of (J_0, J_1, J_2) and σ_b/σ , which the Frison-Pocock parameterization does not deliver
 216 for a slope estimand or for designs with a common close phase.

Table 1: RE of averaged change-score vs GLS on the γ scale ($\text{Var}_{\text{GLS}} / \text{Var}(\hat{\gamma}_{\text{avg}}/c)$) by σ_b/σ .

(J_0, J_1, J_2)	0.5	1	2	5	20
(1,1,0)	0.667	0.500	0.250	0.056	0.004
(2,1,0)	0.653	0.344	0.117	0.021	0.001
(4,1,0)	0.427	0.149	0.041	0.007	0.000
(0,2,0)	0.818	0.900	0.964	0.994	1.000
(1,2,0)	0.529	0.410	0.204	0.045	0.003
(2,2,0)	0.490	0.243	0.080	0.014	0.001
(4,2,0)	0.276	0.089	0.024	0.004	0.000
(0,3,0)	0.771	0.900	0.969	0.995	1.000
(1,3,0)	0.501	0.394	0.193	0.042	0.003
(2,3,0)	0.433	0.209	0.068	0.012	0.001
(4,3,0)	0.220	0.069	0.018	0.003	0.000
(0,1,1)	0.727	0.533	0.245	0.050	0.003
(1,1,1)	0.452	0.258	0.093	0.017	0.001
(2,1,1)	0.422	0.181	0.055	0.009	0.001
(4,1,1)	0.255	0.080	0.021	0.003	0.000
(0,2,1)	0.705	0.517	0.224	0.044	0.003
(1,2,1)	0.448	0.262	0.094	0.017	0.001
(2,2,1)	0.401	0.172	0.052	0.009	0.001
(4,2,1)	0.221	0.068	0.018	0.003	0.000
(0,3,1)	0.685	0.488	0.202	0.039	0.003
(1,3,1)	0.446	0.258	0.092	0.017	0.001
(2,3,1)	0.380	0.162	0.049	0.008	0.001
(4,3,1)	0.194	0.059	0.016	0.003	0.000

Numerically, the ordering holds with zero violations over the 115-cell grid of the tables below. All tables, figures, and simulation results in this paper are on the equal-estimand γ scale: we divide the averaged and ANCOVA contrast variances by c^2 before any comparison. For reference, the raw phase-mean contrasts carry variances larger by the factor c^2 ($c^2 = 2.25$ at the $(2, 2, 0)$ design), so raw-contrast comparisons would overstate the penalty of the simpler estimators.

3.2 When does the gap exceed 5%?

The answer, computed over the full grid of the preceding tables (all on the γ scale), is that this holds in most cells. Of the 115 design-by-ratio cells, 12 have ANCOVA relative efficiency of at least 0.95 (an efficiency loss below 5%), and these form exactly two families. The first (6 cells) comprises single-post-visit run-in designs with no common close ($J_1 = 1, J_2 = 0$): the $(1, 1, 0)$ design at every tabulated ratio, plus $(2, 1, 0)$ at $\sigma_b/\sigma = 0.5$. With a single post-randomization visit, there is no within-period slope information for GLS to exploit. The second (6 cells) comprises designs without run-in or common close ($J_0 = 0, J_2 = 0, J_1 \in \{2, 3\}$) at $\sigma_b/\sigma \geq 2$, where ANCOVA reduces to the averaged estimator and the rescaled phase mean is nearly

Table 2: RE of ANCOVA vs GLS on the γ scale ($\text{Var}_{\text{GLS}} / \text{Var}(\hat{\gamma}_{\text{ANC}}/c)$) by σ_b/σ .

(J_0, J_1, J_2)	0.5	1	2	5	20
(1,1,0)	1.000	1.000	1.000	1.000	1.000
(2,1,0)	0.971	0.909	0.860	0.838	0.834
(4,1,0)	0.838	0.747	0.713	0.702	0.700
(0,2,0)	0.818	0.900	0.964	0.994	1.000
(1,2,0)	0.858	0.866	0.853	0.838	0.834
(2,2,0)	0.808	0.728	0.671	0.648	0.643
(4,2,0)	0.637	0.544	0.512	0.502	0.500
(0,3,0)	0.771	0.900	0.969	0.995	1.000
(1,3,0)	0.808	0.815	0.785	0.758	0.751
(2,3,0)	0.732	0.642	0.582	0.558	0.553
(4,3,0)	0.538	0.450	0.420	0.411	0.409
(0,1,1)	0.727	0.533	0.245	0.050	0.003
(1,1,1)	0.733	0.545	0.390	0.319	0.304
(2,1,1)	0.697	0.542	0.460	0.430	0.425
(4,1,1)	0.587	0.490	0.456	0.446	0.445
(0,2,1)	0.705	0.517	0.224	0.044	0.003
(1,2,1)	0.723	0.541	0.385	0.313	0.299
(2,2,1)	0.678	0.527	0.447	0.418	0.412
(4,2,1)	0.539	0.444	0.413	0.403	0.401
(0,3,1)	0.685	0.488	0.202	0.039	0.003
(1,3,1)	0.699	0.516	0.362	0.293	0.279
(2,3,1)	0.642	0.494	0.416	0.388	0.382
(4,3,1)	0.486	0.397	0.367	0.359	0.357

232 efficient. In run-in designs with $J_1 \geq 2$, the regime that motivates this paper, the loss is
 233 substantial. At the representative $(2, 2, 0)$ design, the relative efficiency ranges from 0.81 at
 234 $\sigma_b/\sigma = 0.5$ down to 0.64 at $\sigma_b/\sigma = 20$, an efficiency loss of 19% to 36%. The averaged
 235 estimator reaches 0.95 only in the second family (6 of 115 cells).

236 4 Robustness to misspecification

237 4.1 Misspecified G

238 When σ_b^2 is misestimated, whether too large or too small, the GLS analysis that as-
 239 sumes it is no longer optimal. Its sampling variance is then the sandwich form
 240 $(X'\Sigma_a^{-1}X)^{-1}X'\Sigma_a^{-1}\Sigma\Sigma_a^{-1}X(X'\Sigma_a^{-1}X)^{-1}$, with Σ_a the assumed and Σ the true covari-
 241 ance. This is the model-robust variance of²⁸ evaluated at a known working covariance rather
 242 than an estimated one. The averaged change-score estimator's definition does not depend
 243 on the assumed value, so its variance stays constant across the sweep. The scale of the
 244 comparison is worth noting: over assumed $\sigma_b^2 \in [0.01, 5]$ with true $\sigma_b^2 = 1$, the misspecified
 245 GLS sandwich variance never exceeds 5.30, which remains below the averaged-estimator
 246 variance of 8.89 on the γ scale. That is, the studied misspecification erodes but does not
 247 overturn the GLS advantage.

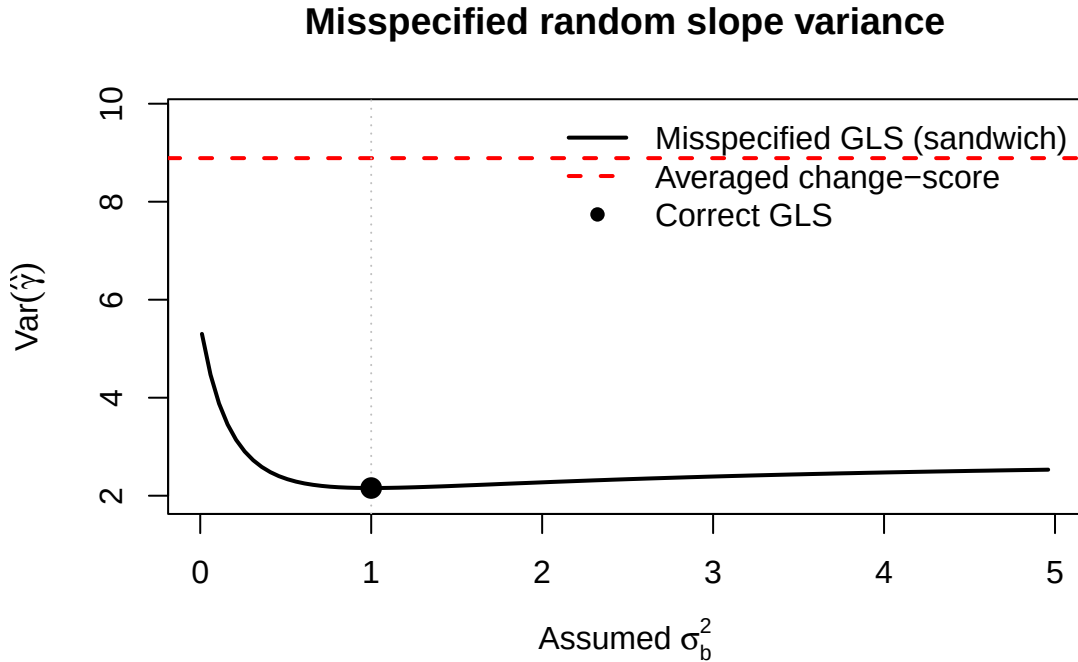


Figure 1: (#fig:misspec_G)Effect of misspecifying σ_b^2 on GLS variance. True $\sigma_b^2 = 1$. The GLS estimator uses an assumed value on the x-axis. The averaged estimator's definition does not use σ_b^2 , so its γ -scale variance (horizontal line) does not depend on the assumed value. The misspecified GLS variance remains below the averaged variance throughout.

248 4.2 Misspecified R

249 When the residual correlation structure is wrong, for example when compound symmetry is
 250 assumed but AR(1) is true, both the GLS variance formula and the Woodbury correction are
 251 biased. The averaged estimator's definition does not depend on the assumed model, though
 252 its γ -scale variance does change with the true R (8.4 to 9.1 across $\rho \in [0, 0.9]$ in the figure).
 253 The GLS sandwich variance ranges from 0.74 to 2.16 over the same sweep. That is, the studied
 254 R misspecification leaves GLS four to eleven times more efficient than the averaged estimator
 255 throughout.

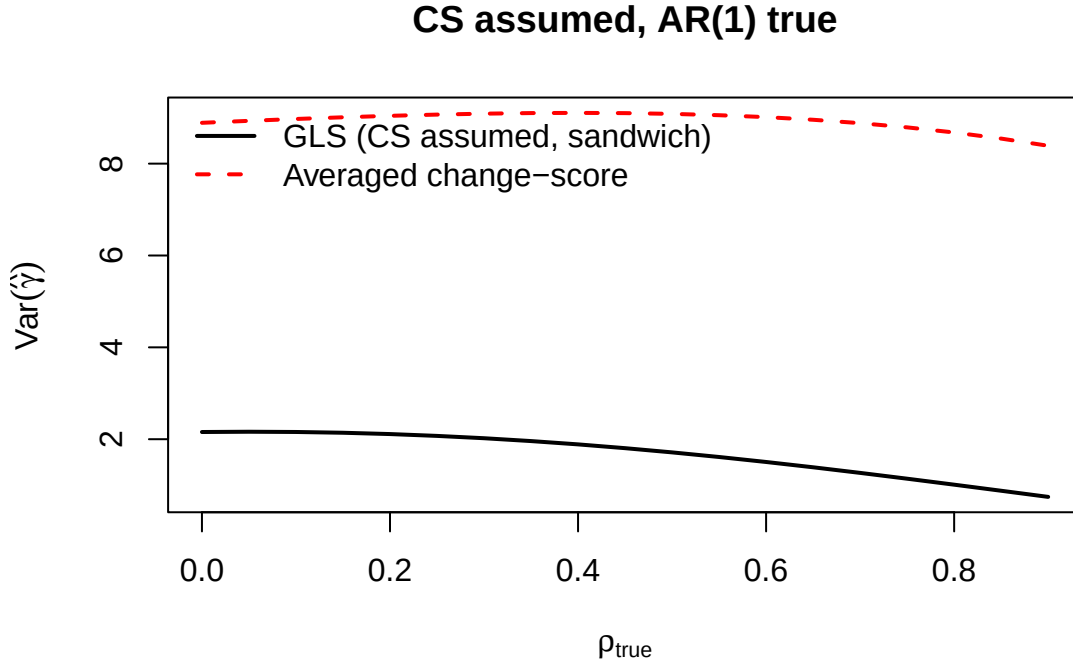


Figure 2: (#fig:misspec_R)Effect of analyzing under compound symmetry when the true correlation is AR(1). Sandwich variance of GLS (solid) vs averaged change-score on the γ scale (dashed) as a function of ρ . The averaged estimator's definition does not use the assumed correlation, but its variance varies with the true R ; GLS remains at least fourfold more efficient at every ρ .

256 5 Simulation study

257 5.1 Design

258 To complement the analytical results, we simulate trials under the following factorial design:

- 259 • $m \in \{50, 100\}$ per group
- 260 • $(J_0, J_1, J_2) \in \{(0, 2, 0), (2, 2, 0), (2, 2, 2)\}$
- 261 • $\sigma_b/\sigma \in \{1, 5\}$
- 262 • True correlation: compound symmetry (an AR(1) truth is examined analytically in
 263 Section 4 but is not simulated)
- 264 • Analysis: GLS (correctly specified), GLS (misspecified), averaged change-score

Table 3: Monte Carlo comparison of GLS (correctly specified), GLS (misspecified σ_b^2 , assuming $\sigma_b^2 = \sigma^2$), and the averaged change-score estimator under compound symmetry (1000 replicates, $\Delta = 0.5$, two-sided $\alpha = 0.05$). Columns report the empirical standard deviation of $\hat{\gamma}$ and the empirical power. The averaged estimator is rescaled to the γ scale (contrast divided by c).

Design	σ_b/σ	m	SD GLS	SD mis	SD avg	Pow GLS	Pow mis	Pow avg
(0,2,0)	1	50	0.238	0.238	0.250	0.537	0.537	0.495
(2,2,0)	1	50	0.210	0.210	0.417	0.677	0.677	0.206
(2,2,2)	1	50	0.151	0.151	0.458	0.902	0.902	0.192
(0,2,0)	1	100	0.169	0.169	0.178	0.823	0.823	0.769
(2,2,0)	1	100	0.146	0.146	0.308	0.926	0.926	0.393
(2,2,2)	1	100	0.108	0.108	0.334	0.997	0.997	0.345
(0,2,0)	5	50	1.012	1.012	1.017	0.083	0.653	0.083
(2,2,0)	5	50	0.241	0.528	2.043	0.555	0.600	0.067
(2,2,2)	5	50	0.164	0.284	2.297	0.873	0.765	0.054
(0,2,0)	5	100	0.726	0.726	0.726	0.108	0.733	0.109
(2,2,0)	5	100	0.162	0.360	1.387	0.857	0.748	0.067
(2,2,2)	5	100	0.106	0.189	1.573	0.997	0.936	0.060

For each configuration, we simulate 1000 trials. The outcomes are the empirical standard deviation of $\hat{\gamma}$ and the empirical power. We do not assess coverage of nominal 95% confidence intervals or empirical size at $\Delta = 0$; extending the factorial to $m \in \{30, 200\}$, $\sigma_b/\sigma = 20$, an AR(1) truth arm, and 5000 replicates remains future work. Test statistics use standard normal critical values rather than t , a distinction that is immaterial at $m \geq 50$ per group.

5.2 Results

We simulate the factorial design described above under compound symmetry, comparing the correctly specified GLS estimator, a GLS estimator that misstates the random-slope variance (it assumes $\sigma_b^2 = \sigma^2$ regardless of the truth), and the averaged change-score estimator, the latter rescaled to the γ scale by dividing the contrast by c . For each cell we report the empirical standard deviation of $\hat{\gamma}$ and the empirical power from 1000 replicates. The Monte Carlo standard error of a tabulated power estimate is at most $\sqrt{0.25/1000} \approx 0.016$.

One feature of the table warrants comment. In the (0,2,0) rows, the correctly specified and misspecified GLS analyses report identical empirical standard deviations but different powers. With $J_0 = 0$, the design has two observations and two fixed effects, so the GLS point estimator does not depend on the assumed covariance, and the two analyses coincide as estimators. The apparent power advantage of the misspecified analysis therefore reflects only an anticonservative model-based standard error, not additional information. More generally, we have not tabulated empirical size at $\Delta = 0$ or interval coverage, so power comparisons across variance models in this table should not be interpreted as efficiency comparisons; adding size and coverage columns remains to be done.

Table 4: Sample size per group: GLS, ANCOVA, and averaged estimators (MIRIAD parameters, $\Delta = 0.25$, 90% power). All variances are on the γ scale (ANCOVA and averaged contrasts divided by c).

(J_0, J_1, J_2)	N GLS	N ANC	N Avg
(4,3,1)	36	95	1630
(4,3,0)	43	100	1676
(4,2,1)	60	144	2413
(2,3,1)	69	162	1053
(4,2,0)	78	151	2368
(2,3,0)	93	157	1028
(1,3,1)	99	255	821
(2,2,1)	104	226	1479
(0,3,1)	112	433	433
(1,2,1)	138	335	1116
(2,2,0)	145	213	1361
(0,2,1)	150	527	527
(4,1,1)	156	339	5328
(1,3,0)	185	233	775
(2,1,1)	226	478	3063

6 Numerical examples

6.1 MIRIAD parameters

6.2 When to use which estimator

A decision rule based on m , σ_b/σ , and confidence in the correlation structure requires the feasible-GLS simulation described in the Discussion, and we defer it to a revision. The results above support only the narrower statement that, with variance components taken as known, GLS dominates, and the penalty of the simpler estimators is largest in run-in designs with $J_1 \geq 2$ and high σ_b/σ .

7 Discussion

The estimator comparison yields a graded recommendation rather than a single winner. Under correct specification, GLS is efficient and dominates both ANCOVA and the averaged change-score estimator, as the ordering $\text{Var}_{\text{GLS}} \leq \text{Var}_{\text{ANC}} \leq \text{Var}_{\text{avg}}$ confirms. We note that this dominance is a large-sample statement: it presumes the covariance components are known or well estimated. With moderate-to-large samples and a credible covariance model, GLS is the appropriate default.

The misspecification analysis qualifies the case for GLS less than might be expected. Because GLS weights the data by an assumed inverse covariance, errors in the assumed G or R feed into its sampling variance, and the sandwich-variance calculations quantify this inflation. In the settings studied, however, the inflation never closes the gap: the misspecified GLS variance

305 peaks at 5.30 in the G sweep and at 2.16 in the R sweep, against γ -scale averaged-estimator
306 variances of 8.4 to 9.1 at the same design. The averaged change-score estimator avoids the
307 variance components entirely, and ANCOVA estimates the run-in coefficient from the data,
308 but under the misspecifications examined, neither robustness property is enough to overcome
309 its efficiency deficit relative to GLS. These calculations take the variance components of the
310 GLS analysis as fixed inputs. The open and practically important question, which we do
311 not answer here, is whether feasible GLS, with components estimated by REML from small
312 samples, keeps its advantage.

313 The practical importance of the estimator choice depends on the design. As the numbers of
314 run-in and treatment observations grow, the phase means approach sufficiency for the subject-
315 specific slope, the ANCOVA efficiency loss vanishes as $J_0 \rightarrow \infty$, and the relative efficiencies
316 of all three estimators converge toward one. The gap is largest, and the choice of estimator
317 therefore most consequential, in small designs with few observations per phase and a high
318 slope-to-noise ratio, which is exactly the regime in which the estimated covariance is also
319 least stable. Whether that instability erodes the feasible-GLS advantage enough to change
320 the ranking in small studies is a question that the planned feasible-GLS simulation must
321 answer; the known-components results presented here cannot.

322 Two estimators not fully developed here would round out the comparison. The seemingly un-
323 related regression formulation of Section 2.4 treats the run-in and treatment periods as linked
324 equations and offers a compromise between the averaged estimator and full GLS, at the cost of
325 additional implementation complexity; its variance derivation remains to be completed. The
326 constrained longitudinal data analysis estimator, on the other hand, uses the full covariance
327 without committing to a random-effects distribution and should fall between ANCOVA and
328 GLS in efficiency. A comprehensive treatment would compare all of these under a feasible-
329 GLS simulation in which the variance components are estimated rather than assumed. That
330 is precisely the setting in which the robustness distinctions drawn here become decisive.

331 8 Conflict of interest

332 The author declares no conflicts of interest.

333 9 Funding

334 This work received no specific funding.

335 10 Data availability statement

336 The `runinpower` R package implementing the GLS, ANCOVA, and averaged-change-score
337 estimators (`ancova.R`, `averaged.R`), the simulation drivers, and the shared bibliography
338 across the four-paper compendium are openly available in the project repository at <https://github.com/rgt47/runinpower>. Specific paths are listed in the Reproducibility section
339 below.
340

341 11 Reproducibility

342 We rendered this manuscript with R 4.4.x. The principal packages used are the in-package
343 `runinpower` source, `tinytest` (testing harness), and `rmarkdown` plus `bookdown` for the
344 manuscript build. The GLS comparator in the simulation is implemented directly in the
345 manuscript source (`analyze_gls` in the simulation chunk); we use no external mixed model
346 fitter.

347 **Random-number generator.** The Monte Carlo simulation comparing the three estima-
348 tors sets `set.seed(20260418)` once, immediately before the replicate loop. We perform no
349 per-replicate reseeding, so the full replicate stream, and hence the tabulated results, follow
350 deterministically from that single seed.

351 **Data and code availability.** The R package source is at `R/`. Simulation drivers
352 are at `analysis/scripts/`. The companion manuscripts (Papers 1, 2, 3) are at
353 `analysis/report/{01,02,03}-*/`. The manuscript source and shared bibliography
354 (`references.bib`, a symlink to `../..references.bib`) are at `analysis/report/04-gls-vs-ttest/`.

355 12 References

- 356 1. Senn S. Change from baseline and analysis of covariance revisited. *Statistics in Medicine*. 2006;25(24):4334-4344. doi:10.1002/sim.2682
- 357 2. Van Breukelen GJP. ANCOVA versus change from baseline had more power in randomized studies and more bias in nonrandomized studies. *Journal of Clinical Epidemiology*. 2006;59(9):920-925. doi:10.1016/j.jclinepi.2006.02.007
- 358 3. Crager MR. Analysis of covariance in parallel-group clinical trials with pretreatment baselines. *Biometrics*. 1987;43(4):895-901. doi:10.2307/2531542
- 359 4. Yang L, Tsiatis AA. Efficiency study of estimators for a treatment effect in a pretest-posttest trial. *The American Statistician*. 2001;55(4):314-321. doi:10.1198/000313001753272466
- 360 5. Frison L, Pocock SJ. Repeated measures in clinical trials: Analysis using mean summary statistics and its implications for design. *Statistics in Medicine*. 1992;11(13):1685-1704. doi:10.1002/sim.4780111304
- 361 6. Frison L, Pocock SJ. Linearly divergent treatment effects in clinical trials with repeated measures: Efficient analysis using summary statistics. *Statistics in Medicine*. 1997;16(24):2855-2872. doi:10.1002/(SICI)1097-0258(19971230)16:24<2855::AID-SIM749>3.0.CO;2-2

- 362 7. Hu N, Mackey H, Thomas R. Power and sample size for random coefficient regression models in randomized experiments with monotone missing data. *Biometrical Journal*. 2021;63(4):806-824. doi:10.1002/bimj.202000184
- 363 8. Liu GF, Lu K, Mogg R, Mallick M, Mehrotra DV. Should baseline be a covariate or dependent variable in analyses of change from baseline in clinical trials? *Statistics in Medicine*. 2009;28(20):2509-2530. doi:10.1002/sim.3639
- 364 9. Lu K. On efficiency of constrained longitudinal data analysis versus longitudinal analysis of covariance. *Biometrics*. 2010;66(3):891-896. doi:10.1111/j.1541-0420.2009.01322.x
- 365 10. Tsiatis AA, Davidian M, Zhang M, Lu X. Covariate adjustment for two-sample treatment comparisons in randomized clinical trials: A principled yet flexible approach. *Statistics in Medicine*. 2008;27(23):4658-4677. doi:10.1002/sim.3113
- 366 11. Mallinckrodt CH, Ye J, et al. Mixed models for repeated measures should include time-by-covariate interactions to assure power gains and robustness against dropout bias relative to complete-case ANCOVA. *Statistics in Biopharmaceutical Research*. Published online 2021. doi:10.1080/19466315.2021.1976846
- 367 12. Brogan DR, Kutner MH. Comparative analyses of pretest-posttest research designs. *The American Statistician*. 1980;34(4):229-232. doi:10.1080/00031305.1980.10483034
- 368 13. Coffman CJ, Edelman D, Woolson RF. To condition or not condition? Analysing “change” in longitudinal randomised controlled trials. *BMJ Open*. 2016;6(12):e013096. doi:10.1136/bmjopen-2016-013096
- 369 14. Wang G, Aschenbrenner AJ, Li Y, et al. Two-period linear mixed effects models to analyze clinical trials with run-in data when the primary outcome is continuous: Applications to Alzheimer’s disease. *Alzheimer’s & Dementia: Translational Research & Clinical Interventions*. 2019;5:450-457. doi:10.1016/j.trci.2019.08.002
- 370 15. Liang KY, Zeger SL. Longitudinal data analysis of continuous and discrete responses for pre-post designs. *Sankhya: The Indian Journal of Statistics, Series B*. 2000;62(1):134-148.
- 371 16. Funatogawa I, Funatogawa T. Longitudinal analysis of pre- and post-treatment measurements with equal baseline assumptions in randomized trials. *Biometrical Journal*. 2020;62(5):1235-1249. doi:10.1002/bimj.201800389

- 372 17. Iddi S, Donohue MC. Power and sample size for longitudinal models in R – the long-power package and Shiny app. *The R Journal*. 2022;14(1):264-282. <https://journal.r-project.org/articles/RJ-2022-022/>
- 373 18. Liu G, Liang KY. Sample size calculations for studies with correlated observations. *Biometrics*. 1997;53:937-947. doi:10.2307/2533554
- 374 19. Diggle PJ, Heagerty P, Liang KY, Zeger SL. *Analysis of Longitudinal Data*. 2nd ed. Oxford University Press; 2002.
- 375 20. Ard MC, Edland SD. Power calculations for clinical trials in Alzheimer’s disease. *Journal of Alzheimer’s Disease*. 2011;21:369-377. doi:10.3233/JAD-2011-0062
- 376 21. Frost C, Kenward MG, Fox NC. Optimizing the design of clinical trials where the outcome is a rate. Can estimating a baseline rate in a run-in period increase efficiency? *Statistics in Medicine*. 2008;27(18):3717-3731. doi:10.1002/sim.3280
- 377 22. Laird NM, Ware JH. Random-effects models for longitudinal data. *Biometrics*. 1982;38(4):963-974. doi:10.2307/2529876
- 378 23. Verbeke G, Molenberghs G. *Linear Mixed Models for Longitudinal Data*. Springer; 2000.
- 379 24. Borm GF, Fransen J, Lemmens WAJG. A simple sample size formula for analysis of covariance in randomized clinical trials. *Journal of Clinical Epidemiology*. 2007;60(12):1234-1238. doi:10.1016/j.jclinepi.2007.02.006
- 380 25. Vickers AJ. The use of percentage change from baseline as an outcome in a controlled trial is statistically inefficient: A simulation study. *BMC Medical Research Methodology*. 2001;1:6. doi:10.1186/1471-2288-1-6
- 381 26. Henderson CR. Best linear unbiased estimation and prediction under a selection model. *Biometrics*. 1975;31(2):423-447. doi:10.2307/2529430
- 382 27. Robinson GK. That BLUP is a good thing: The estimation of random effects. *Statistical Science*. 1991;6(1):15-32. doi:10.1214/ss/1177011926
- 383 28. Liang KY, Zeger SL. Longitudinal data analysis using generalized linear models. *Biometrika*. 1986;73(1):13-22. doi:10.1093/biomet/73.1.13